CHEMISTRY · GRADE 10

Ionic Compounds and Metals

A study guide for Lesson 3 (Names and Formulas for Ionic Compounds) and Lesson 4 (Metallic Bonds and the Properties of Metals), with practice questions and answers

Contents

Lesson 3: Names and Formulas for Ionic Compounds

1. Formulas for ionic compounds and the formula unit
2. Monatomic ions and oxidation numbers
3. Writing formulas for binary ionic compounds
4. Polyatomic ions and oxyanions
5. Naming ionic compounds
6. Practice set A

Lesson 4: Metallic Bonds and the Properties of Metals

7. Metallic bonds and the electron sea model
8. Comparing metals and ionic compounds
9. Properties of metals
0. Metal alloys
1. Practice set B

Chapter review and answer key

How to use this guide: read each section, study the worked examples, then try the practice set before checking the answer key at the end.

LESSON 3

NAMES AND FORMULAS FOR IONIC COMPOUNDS

FOCUS QUESTION

What are the names and formulas of ionic compounds?

Formulas for Ionic Compounds

Chemists around the world need to communicate clearly with one another, so they use one standard set of rules for naming compounds. Once you know these rules, you can write a chemical formula from a compound's name, and name a compound from its formula.

An ionic compound is not made of separate molecules. It is a huge crystal of positive and negative ions arranged in a repeating pattern. Because there is no single molecule to describe, chemists use the **formula unit**: the chemical formula for an ionic compound, which represents the *simplest ratio* of the ions involved.

> **Example: magnesium chloride, $MgCl_2$**
> The formula unit is $MgCl_2$ because magnesium ions (Mg^{2+}) and chloride ions (Cl^-) exist in a **1 : 2 ratio**. One Mg^{2+} (+2) is balanced by two Cl^- (−1 each), so the total charge is zero.

KEY IDEA

The total charge of a formula unit is always **zero**. The positive charge of the cations must exactly cancel the negative charge of the anions.

Monatomic Ions

A **monatomic ion** is an ion made from a single atom, such as Na^+, Ca^{2+} or O^{2-}. For main-group elements, the charge can be predicted from the element's group in the periodic table, because atoms gain or lose electrons to reach a stable octet (noble-gas configuration).

Group	Electrons gained/lost	Ion charge	Examples
1	loses 1	1+	Li^+, Na^+, K^+
2	loses 2	2+	Mg^{2+}, Ca^{2+}, Ba^{2+}
13	loses 3	3+	Al^{3+}, Ga^{3+}
15	gains 3	3−	N^{3-} (nitride), P^{3-} (phosphide)
16	gains 2	2−	O^{2-} (oxide), S^{2-} (sulfide)
17	gains 1	1−	F^- (fluoride), Cl^- (chloride), Br^-, I^-

Transition metals (groups 3–12) often form *more than one* ion. Iron, for example, forms Fe^{2+} and Fe^{3+}; copper forms Cu^+ and Cu^{2+}. Some, like silver (Ag^+) and zinc (Zn^{2+}), form only one common ion.

Oxidation number

The charge of a monatomic ion is called its **oxidation number** (or oxidation state). The oxidation number of Na^+ is +1, of O^{2-} is −2, and of Fe^{3+} is +3. In an ionic formula, the sum of all oxidation numbers equals zero.

Writing Formulas for Binary Ionic Compounds

A **binary ionic compound** contains only two elements: a metal cation and a nonmetal anion. Follow these steps:

1. Write the symbol of the **cation first**, then the anion.
2. Write each ion's charge (oxidation number).
3. Add **subscripts** so that the total charge is zero. A quick method is the *crisscross method*: the number of each charge becomes the subscript of the other ion.
4. Reduce the subscripts to the simplest whole-number ratio. Never write a subscript of 1.

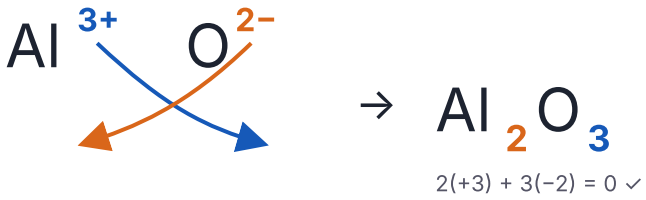


Figure 1. The crisscross method for aluminum oxide.

Worked example 1: calcium fluoride
Ca^{2+} and F^-. Two F^- ions are needed to balance one Ca^{2+}.
Formula: **CaF_2** Check: (+2) + 2(−1) = 0

Worked example 2: magnesium oxide
Mg^{2+} and O^{2-}. Crisscross gives Mg_2O_2, which reduces to 1 : 1.
Formula: **MgO** Check: (+2) + (−2) = 0

Worked example 3: iron(III) chloride
The Roman numeral tells you the ion is Fe^{3+}. With Cl^-:
Formula: **$FeCl_3$** Check: (+3) + 3(−1) = 0

Worked example 4: sodium sulfide
Na^+ and S^{2-}. Two Na^+ balance one S^{2-}.
Formula: **Na_2S** Check: 2(+1) + (−2) = 0

Polyatomic Ions

A **polyatomic ion** is a group of covalently bonded atoms that carries an overall charge and behaves as a single unit, for example the sulfate ion, SO_4^{2-}. The charge belongs to the whole group, not to any one atom.

Name	Formula	Name	Formula
ammonium	NH_4^+	sulfate	SO_4^{2-}
hydroxide	OH^-	sulfite	SO_3^{2-}
nitrate	NO_3^-	carbonate	CO_3^{2-}
nitrite	NO_2^-	chromate	CrO_4^{2-}
hydrogen carbonate	HCO_3^-	dichromate	$Cr_2O_7^{2-}$
acetate	CH_3COO^-	peroxide	O_2^{2-}
cyanide	CN^-	phosphate	PO_4^{3-}
permanganate	MnO_4^-	hydrogen phosphate	HPO_4^{2-}

Formulas with polyatomic ions

Use exactly the same charge-balancing rule. If you need **more than one** polyatomic ion, put it in **parentheses** and write the subscript outside. Never change the subscripts inside a polyatomic ion.

Calcium hydroxide
$Ca^{2+} + 2\,OH^- \rightarrow \mathbf{Ca(OH)_2}$

Ammonium phosphate
$3\,NH_4^+ + PO_4^{3-} \rightarrow \mathbf{(NH_4)_3PO_4}$

COMMON MISTAKE

Writing $CaOH_2$ instead of $Ca(OH)_2$. Without parentheses, the 2 applies only to H, which is a different (and wrong) formula. When only one polyatomic ion is needed, as in $NaNO_3$, no parentheses are used.

Oxyanions

An **oxyanion** is a polyatomic ion made of an element (usually a nonmetal) bonded to one or more oxygen atoms. Many elements form a family of oxyanions. Their names follow a pattern:

- The ion with **more** oxygen atoms ends in **-ate**; the ion with **fewer** ends in **-ite** (nitrate NO_3^-, nitrite NO_2^-).
- For halogens with four oxyanions, add the prefix **per-** (one more O than -ate) and **hypo-** (one fewer O than -ite).

Ion	Name	Number of O atoms
ClO_4^-	**per**chlor**ate**	4 (most)
ClO_3^-	chlor**ate**	3
ClO_2^-	chlor**ite**	2
ClO^-	**hypo**chlor**ite**	1 (fewest)

Naming Ionic Compounds

To name an ionic compound from its formula, apply these rules in order:

1. **Name the cation first, then the anion.** NaCl is sodium chloride, not chloride sodium.
2. **A monatomic cation takes the name of the element.** Ca^{2+} is the calcium ion.
3. **A monatomic anion takes the root of the element name plus *-ide*.** Br^- is bromide; N^{3-} is nitride.
4. **If the metal can form more than one ion**, show its charge with a Roman numeral in parentheses (the *Stock system*). Fe^{2+} = iron(II); Fe^{3+} = iron(III).
5. **If the compound contains a polyatomic ion**, simply use the polyatomic ion's name. NaOH is sodium hydroxide.

Worked example 5: name CuO

Copper can be Cu^+ or Cu^{2+}, so find its charge. Oxide is O^{2-}, and there is one O for one Cu, so copper must be +2.
Name: **copper(II) oxide**

Worked example 6: name $Fe_2(SO_4)_3$

Three sulfates give $3 \times (-2) = -6$. Two Fe ions must give +6, so each Fe is +3.
Name: **iron(III) sulfate**

Worked example 7: name K_2CrO_4 and NH_4Cl

Potassium has only one ion (K^+), so no Roman numeral: **potassium chromate**.
NH_4^+ is ammonium: **ammonium chloride**.

SUMMARY FLOWCHART

Is the cation a metal with more than one possible charge (most transition metals, also Sn and Pb)? → **Yes:** add a Roman numeral. **No** (groups 1, 2, Al, Zn, Ag): no Roman numeral.
Is the anion a single element? → **Yes:** use *-ide*. **No:** use the polyatomic ion's name.

Practice Set A: Names and Formulas

Part 1: Write the formula.

1. potassium iodide
2. barium chloride
3. aluminum sulfide
4. lithium nitride
5. copper(I) oxide
6. lead(IV) oxide
7. magnesium nitrate
8. sodium carbonate
9. aluminum hydroxide
10. ammonium sulfate

Part 2: Name the compound.

11. CaO
12. Na_3P
13. $FeCl_2$
14. $CuSO_4$
15. $KMnO_4$
16. $Ca(ClO)_2$
17. $Ba(OH)_2$
18. Fe_2O_3

Part 3: Think about it.

19. Explain why the formula of an ionic compound is called a *formula unit* rather than a molecular formula.
20. A student writes the formula for calcium nitrate as $CaNO_32$. Identify the mistake and write the correct formula.
21. Why does the name *sodium chloride* not need a Roman numeral, while *iron(III) chloride* does?
22. Name the four oxyanions of bromine: BrO_4^-, BrO_3^-, BrO_2^-, BrO^-.

LESSON 4

METALLIC BONDS AND THE PROPERTIES OF METALS

FOCUS QUESTION

How do metals and ionic compounds compare and contrast?

Metallic Bonds

Although metals are not ionic, they share several properties with ionic compounds. In both cases, bonding is based on the **attraction of particles with unlike charges**. In the solid state, metals often form lattices similar to ionic crystal lattices. In such a lattice, each metal atom is surrounded by **8 to 12** other metal atoms.

The electron sea model

Metal atoms have few valence electrons, and they hold them loosely. In a metal lattice, the outer energy levels of neighbouring atoms overlap. The valence electrons are therefore not tied to any one atom; they move freely through the whole solid. The **electron sea model** describes a metal as a lattice of positive metal ions (cations) immersed in a "sea" of moving valence electrons.

Electrons that are free to move and are not held by any specific atom are called **delocalized electrons**. A **metallic bond** is the attraction of a metallic cation for delocalized electrons.

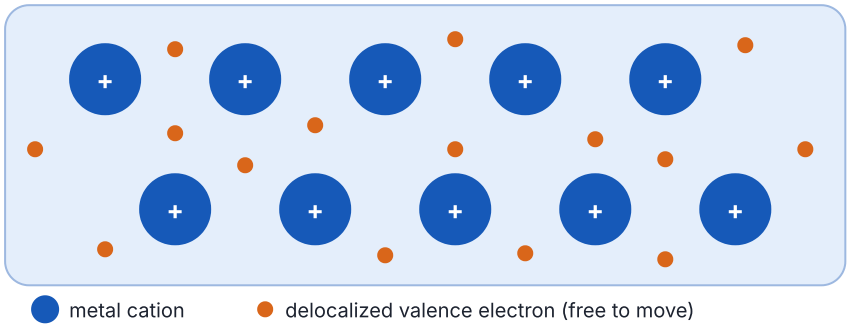


Figure 2. The electron sea model: fixed metal cations surrounded by mobile, delocalized electrons.

Comparing Metals and Ionic Compounds

Feature	Ionic compounds	Metals
Particles	Cations and anions	Cations and delocalized electrons
Bond	Electrostatic attraction between oppositely charged ions	Attraction between cations and the electron sea
Structure	Crystal lattice	Lattice; each atom has 8–12 neighbours
Electrons	Transferred and held by the anions	Shared by all atoms, free to move
Melting point	High	Moderate to very high (varies a lot)
Conductivity (solid)	Does not conduct (ions fixed)	Conducts well (electrons move)
Conductivity (molten/dissolved)	Conducts (ions move)	Conducts
When hammered	Brittle: shatters	Malleable: bends and flattens

Properties of Metals

The physical properties of metals can be explained by metallic bonding. These properties also give evidence about the strength of metallic bonds.

Melting and boiling points

The melting points of metals vary greatly. Mercury is a liquid at room temperature (melting point −39 °C), which makes it useful in scientific instruments such as older thermometers and barometers. Gallium melts at about 30 °C, so it can melt in your hand. Tungsten has the highest melting point of any metal, 3422 °C, so it is used in light-bulb filaments.

Boiling points of metals tend to be **extremely high**. Melting does not require much energy to move cations and electrons past each other; the metal stays bonded as a liquid. But boiling requires each atom to be separated completely from the electron sea and the other cations, which needs a great deal of energy. For example, copper melts at 1085 °C but boils at 2562 °C.

Malleability, ductility and durability

Metals are **malleable** (can be hammered into sheets) and **ductile** (can be drawn into wires). When a force is applied, the cations slide past each other, but the electron sea moves with them and keeps them bonded. The metal changes shape instead of breaking. In an ionic crystal, sliding brings like charges next to each other; they repel, and the crystal shatters.

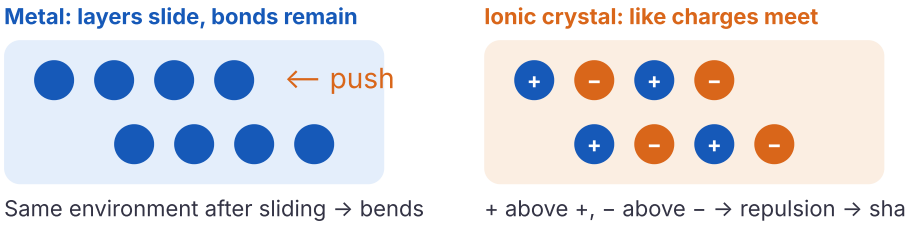


Figure 3. After a layer is shifted, a metal keeps its bonding, but an ionic crystal lines up like charges.

Metals are generally **durable**. Their cations are strongly attracted to the delocalized electrons and are not easily removed from the metal.

Thermal and electrical conductivity

Delocalized electrons move easily through the metal. They carry electric current when a voltage is applied, and they carry heat quickly from one place to another. This is why copper is used in electrical wiring and why metal pans heat up quickly.

Luster

Delocalized electrons interact with light: they absorb and release photons. This gives metals their shiny appearance, called **luster**.

Hardness and strength

The more delocalized electrons there are, the stronger the metallic bonds. Alkali metals (group 1) have only one valence electron per atom, so they are soft enough to cut with a knife. Transition metals such as chromium, iron and nickel can delocalize both their outer *s* electrons and some inner *d* electrons. This produces a denser electron sea, so these metals are hard and strong. Chromium is one of the hardest elements.

REMEMBER

Almost every property of metals (conductivity, malleability, ductility, luster, high boiling point) comes from one idea: **mobile, delocalized electrons** holding a lattice of cations together.

Metal Alloys

Because of the nature of metallic bonding, it is easy to add other elements to a metal. An **alloy** is a mixture of elements that has metallic properties. Alloys are made by melting a metal and mixing in the other elements, then letting the mixture cool. Alloys often have properties that are better than those of the pure metal; for example, they can be harder, stronger or more resistant to corrosion.

SUBSTITUTIONAL ALLOY

Some atoms of the original metal are **replaced** by other atoms of **similar size**.
Examples: brass (copper + zinc), sterling silver (silver + copper), 14-karat gold.

INTERSTITIAL ALLOY

Small atoms fit into the **holes (interstices)** between the larger metal atoms.
Example: carbon steel, where small carbon atoms sit between iron atoms, making it harder and stronger than pure iron.

Alloy	Composition	Uses
Brass	Copper and zinc	Musical instruments, plumbing fittings, hardware
Bronze	Copper and tin	Statues, medals, bells
Stainless steel	Iron, chromium, nickel, carbon	Cutlery, kitchen sinks, surgical instruments
Carbon steel	Iron and carbon	Buildings, bridges, car bodies
Sterling silver	Silver and copper	Jewellery, tableware
Solder	Tin and lead (or tin and copper)	Joining electrical components

Practice Set B: Metallic Bonds and Properties

1. Define **metallic bond** and **delocalized electrons**.
2. Describe the electron sea model in your own words. Draw a labelled sketch.
3. State two similarities and two differences between metallic bonding and ionic bonding.
4. Explain why metals are malleable but ionic crystals are brittle.
5. Explain why the boiling points of metals are much higher than their melting points.
6. Why does a solid metal conduct electricity, while a solid ionic compound does not?
7. Sodium can be cut with a knife, but chromium is very hard. Explain this difference using the number of delocalized electrons.
8. Classify each alloy as substitutional or interstitial: (a) brass (b) carbon steel (c) sterling silver.
9. Why do engineers often use alloys instead of pure metals? Give one example.
10. Mercury and tungsten are both metals. Suggest one use for each based on its melting point.

CHAPTER REVIEW

MIXED PRACTICE

Section 1: Multiple Choice

1. What is the correct formula for aluminum oxide?
   A. AlO B. Al_3O_2 C. Al_2O_3 D. AlO_3
2. Which is the correct name for $CuCl_2$?
   A. copper chloride B. copper(I) chloride C. copper(II) chloride D. copper dichloride

3. Which ion has the most oxygen atoms?
   A. chlorite B. hypochlorite C. chlorate D. perchlorate

4. The formula unit of magnesium chloride shows the ions in a ratio of:
   A. 1 : 1 B. 1 : 2 C. 2 : 1 D. 2 : 3

5. In a metal lattice, each atom is typically surrounded by:
   A. 2–4 atoms B. 4–6 atoms C. 8–12 atoms D. 20 atoms

6. Which property of metals is mainly due to delocalized electrons interacting with light?
   A. luster B. ductility C. hardness D. density

7. Which formula is written correctly?
   A. $Mg(OH)_2$ B. $MgOH_2$ C. Mg_2OH D. $Mg(OH)$

8. Carbon steel is an example of:
   A. an ionic compound B. a substitutional alloy C. an interstitial alloy D. a pure metal

Section 2: Complete the Table

Cation	Anion	Formula	Name
K^+	S^{2-}	_____	_____
Ca^{2+}	PO_4^{3-}	_____	_____
Fe^{3+}	NO_3^-	_____	_____
NH_4^+	CO_3^{2-}	_____	_____
Sn^{4+}	O^{2-}	_____	_____
Al^{3+}	SO_4^{2-}	_____	_____

Section 3: Extended Response

1. **Focus question.** In a short paragraph, answer: *How do metals and ionic compounds compare and contrast?* Include structure, bonding, conductivity and response to force.
2. A sample of an unknown solid is shiny, bends when hammered and conducts electricity as a solid. A second solid is white, shatters when hammered and conducts only when dissolved in water. Identify which is a metal and which is an ionic compound, and justify your answer.
3. Determine the charge on the metal ion in Mn_2O_7, then name the compound.

ANSWER KEY

CHECK YOUR WORK

Practice Set A

1. KI
2. $BaCl_2$
3. Al_2S_3
4. Li_3N
5. Cu_2O
6. PbO_2
7. $Mg(NO_3)_2$
8. Na_2CO_3
9. $Al(OH)_3$
10. $(NH_4)_2SO_4$
11. calcium oxide
12. sodium phosphide
13. iron(II) chloride
14. copper(II) sulfate
15. potassium permanganate
16. calcium hypochlorite
17. barium hydroxide
18. iron(III) oxide
19. An ionic compound is a continuous lattice of ions, not separate molecules. The formula unit only shows the simplest ratio of the ions.
20. The nitrate ion must be in parentheses with the subscript outside: $Ca(NO_3)_2$.
21. Sodium forms only one ion (Na^+), so its charge is never in doubt. Iron can form Fe^{2+} or Fe^{3+}, so the Roman numeral is needed to show which.
22. perbromate, bromate, bromite, hypobromite

Practice Set B

1. Metallic bond: the attraction of a metallic cation for delocalized electrons. Delocalized electrons: electrons that move freely through the metal and are not held by any specific atom.
2. Metal cations sit in a fixed lattice surrounded by a "sea" of mobile valence electrons that belong to the whole metal.
3. Similar: both involve attraction between opposite charges and both form lattices. Different: ionic bonds have anions while metals have free electrons; metals conduct as solids and are malleable, but ionic solids do not conduct and are brittle.
4. In metals, cations slide but the electron sea keeps them bonded. In ionic crystals, sliding puts like charges together, and the repulsion shatters the crystal.
5. Melting only lets particles move past each other while still bonded. Boiling requires atoms to be pulled completely away from the electron sea, which needs much more energy.
6. A metal has mobile electrons to carry charge. In a solid ionic compound, the ions are locked in place.
7. Sodium has one valence electron per atom, so it has a thin electron sea and weak bonds. Chromium delocalizes *s* and *d* electrons, which gives stronger bonding.
8. (a) substitutional (b) interstitial (c) substitutional
9. Alloys can be harder, stronger or more corrosion-resistant. For example, stainless steel does not rust like iron.
10. Mercury: thermometers or barometers, because it is liquid at room temperature. Tungsten: light-bulb filaments, because it withstands very high temperatures.

Chapter Review

Section 1: 1 C · 2 C · 3 D · 4 B · 5 C · 6 A · 7 A · 8 C

Section 2: K_2S, potassium sulfide · $Ca_3(PO_4)_2$, calcium phosphate · $Fe(NO_3)_3$, iron(III) nitrate · $(NH_4)_2CO_3$, ammonium carbonate · SnO_2, tin(IV) oxide · $Al_2(SO_4)_3$, aluminum sulfate

Section 3: (1) Both are lattices held together by attraction of opposite charges. Ionic compounds contain cations and anions, are brittle, and conduct only when molten or dissolved. Metals contain cations in a sea of delocalized electrons, are malleable and ductile, and conduct as solids. (2) The first solid is the metal (luster, malleable, conducts as a solid). The second is ionic (brittle, conducts only when its ions are free to move in solution). (3) Seven O^{2-} give −14, so two Mn give +14 and each Mn is +7: manganese(VII) oxide.